G H Raisoni International Skill Tech University, Pune

Established by Maharashtra Act No. IX of 2024 and under section 2(f) of the UGC Act 1956

End Semester Examinations - Summer Term 2025

School of Engineering and Technology

S.Y. B.Tech., Semester - III (CE/IT/CS/AI/AIML/AIDS/DS)

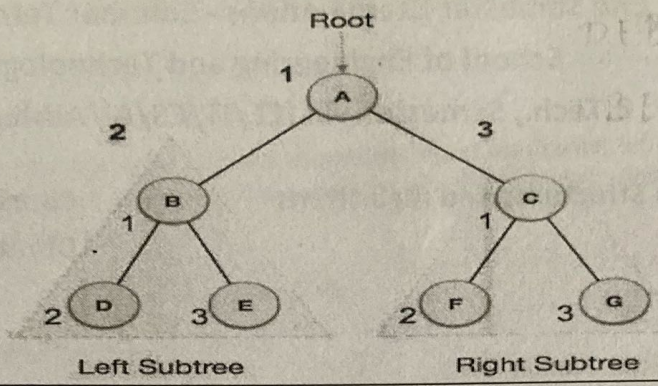
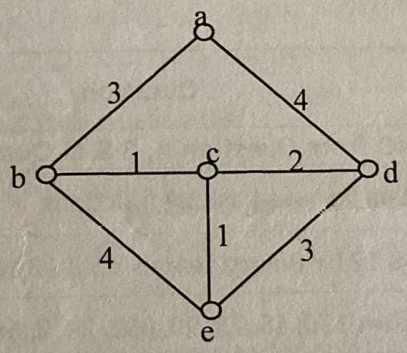
Course Name: Data Structures and Algorithms | **Course Code: 25BTPCL302**

Max Marks: 60 | **Duration: 150 minutes**

Instructions:

1. Write the solution in detail.
2. Clearly show all steps of your work, including necessary derivations and calculations.
3. Read each question carefully before attempting.
4. Students must label their answers with the corresponding question number, such as Q. No. 1A, Q. No. 2C, etc.
5. Each question is of 12 marks, from every question solve any **TWO from A, B & C**. Question **D is compulsory**.

Q. No.	Sub Q.No.	Question	Marks	CO	BL
1		**Solve any TWO from Question A, B & C. Question D is compulsory.**			
	A	State and explain various hashing functions	5	I	2
	B	Search element 12 by Binary Search 2 5 8 12 16 23 38 56 72 91	5	I	3
	C	Sort by Radix Sort 170, 45, 75, 90, 802, 24, 2, 66	5	I	2
	D	Determine and explain various types of data structures	2	I	3
2		**Solve any TWO from Question A, B & C. Question D is compulsory.**			
	A	Compare Array and Linked List	5	II	2
	B	Analyze how we can represent Stack using Linked List	5	II	3
	C	Illustrate the concept of array with its operations	5	II	2
	D	What are the Operations on Linked List	2	II	2
3		**Solve any TWO from Question A, B & C. Question D is compulsory.**			
	A	Differentiate between stack and queue	5	III	2
	B	Convert Infix to Postfix (A+B/C−(D*E)+G) ABC/+DE*-G+	5	III	3
	C	Solve Tower of Hanoi for Disks =3	5	III	3
	D	Define a stack. What are the basic operations of a stack?	2	III	1

4		**Solve any TWO from Question A, B & C. Question D is compulsory.**			
	A	Construct Binary Search Tree from given data 10, 5, 1, 7, 40, 50	5	IV	3
	B	Sort the Data 4, 3, 7, 1, 8, 5 using Heap Sort	5	IV	3
	C	Traverse the tree by Preorder, Postorder and Inorder 	5	IV	3
	D	Explain degree of node with example	2	IV	2
5		**Solve any TWO from Question A, B & C. Question D is compulsory.**			
	A	Find Minimum Spanning Tree by Kruskal's Algorithm 	5	V	3
	B	Explain Various operations on Graph.	5	V	2
	C	Illustrate the concepts of complete graph and regular graph	5	V	2
	D	Draw Biaprtite Graph for A3,2	2	V	3

- End of the Question Paper -